Earthquake Prediction Model using Python

Project Title: Earthquake Prediction

Phase 5: Final submission

1. Introduction:

Earthquakes pose significant threats to human safety and infrastructure. This project aims to develop a predictive model for earthquake occurrences using Python. The motivation behind this project is to provide a tool that can potentially contribute to early warning systems.

2. Project Overview:

This project involves the creation of a machine learning model to predict the likelihood of earthquakes based on historical seismic data. The scope includes data collection, preprocessing, model development, and evaluation.

3. Tools and Technologies:

Programming Language: Python
Libraries: NumPy, Pandas, Scikit-learn, TensorFlow
IDE: Jupyter Notebook

4. Dataset:

The earthquake dataset was obtained from [source]. It comprises [number] instances with features such as [list of features]. Data preprocessing involved handling missing values, scaling, and encoding categorical variables.

5. Data Preprocessing:

Data cleaning included removing duplicates, handling outliers, and normalizing numerical features. Categorical variables were encoded, and missing values were imputed using [method]. Exploratory Data Analysis (EDA) visualizations are available in the Jupyter Notebook.

6. Model Architecture:

The earthquake prediction model utilizes a [type of model] architecture. The choice was made due to its effectiveness in capturing temporal patterns in seismic data. The model consists of [number] layers with [activation functions] and was trained using [optimizer].

7. Model Training:

The model was trained on [percentage]% of the dataset, with the remaining [percentage]% reserved for testing. Hyperparameter tuning involved [parameters], and the training process achieved [performance metrics]. Cross-validation was implemented to ensure robustness.

8. Results:

Evaluation metrics, including accuracy, precision, recall, and F1 score, are presented in the Jupyter Notebook. Visualizations such as confusion matrices and ROC curves provide insights into the model's performance. The model demonstrated [strengths] and exhibited [limitations].

Importing Libraries and Dataset :

Python libraries make it very easy for us to handle the data and perform typical and complex tasks with a single line of code.

- **Pandas** – This library helps to load the data frame in a 2D array format and has multiple functions to perform analysis tasks in one go.
- **Matplotlib/Seaborn** – This library is used to draw visualizations.

Program :

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sb

import warnings
warnings.filterwarnings('ignore')
```

Now, let’s load the dataset into the panda’s data frame for easy analysis.

```
df = pd.read_csv('dataset.csv')
df.head()
```

OUTPUT :

	Origin Time	Latitude	Longitude	Depth	Magnitude	Location
0	2021-07-31 09:43:23 IST	29.06	77.42	5.0	2.5	53km NNE of New Delhi, India
1	2021-07-30 23:04:57 IST	19.93	72.92	5.0	2.4	91km W of Nashik, Maharashtra, India
2	2021-07-30 21:31:10 IST	31.50	74.37	33.0	3.4	49km WSW of Amritsar, Punjab, India
3	2021-07-30 13:56:31 IST	28.34	76.23	5.0	3.1	50km SW of Jhajjar, Haryana
4	2021-07-30 07:19:38 IST	27.09	89.97	10.0	2.1	53km SE of Thimphu, Bhutan

The dataset we are using here contains data for the following columns:

- Origin time of the Earthquake
- Latitude and the longitude of the location.
- Depth – This means how much depth below the earth's level the earthquake started.
- The magnitude of the earthquake
- Location

```
df.shape
```

OUTPUT :

```
(2719, 6)
```

Now let's see which data is present in which type of data format.

```
df.info()
```

OUTPUT :

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2719 entries, 0 to 2718
Data columns (total 6 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Origin Time  2719 non-null   object 
 1   Latitude     2719 non-null   float64
 2   Longitude    2719 non-null   float64
 3   Depth        2719 non-null   float64
 4   Magnitude    2719 non-null   float64
 5   Location     2719 non-null   object 
dtypes: float64(4), object(2)
memory usage: 127.6+ KB
```

Looking at the descriptive statistical measures also gives us some idea regarding the distribution of the data.

```
df.describe()
```

OUTPUT :

	Latitude	Longitude	Depth	Magnitude
count	2719.000000	2719.000000	2719.000000	2719.000000
mean	29.939433	80.905638	53.400478	3.772196
std	7.361564	10.139075	68.239737	0.768076
min	0.120000	60.300000	0.800000	1.500000
25%	25.700000	71.810000	10.000000	3.200000
50%	31.210000	76.610000	15.000000	3.900000
75%	36.390000	92.515000	82.000000	4.300000
max	40.000000	99.960000	471.000000	7.000000

From the above description of the dataset, we can conclude that:

- The maximum magnitude of the Earthquake is 7.
- The maximum depth at which the earthquake started is 471 km below the ground.

Feature Engineering :

Feature Engineering helps to derive some valuable features from the existing ones. These extra features sometimes help in increasing the performance of the model significantly and certainly help to gain deeper insights into the data.

```
splitted = df['Origin Time'].str.split(' ', n=1,
                                        expand=True)

df['Date'] = splitted[0]
df['Time'] = splitted[1].str[:-4]

df.drop('Origin Time',
        axis=1,
        inplace=True)
df.head()
```

OUTPUT :
Now, let’s divide the date column into the day, month, and year columns respectively.

	Latitude	Longitude	Depth	Magnitude	Location	Date	Time
0	29.06	77.42	5.0	2.5	53km NNE of New Delhi, India	2021-07-31	09:43:23
1	19.93	72.92	5.0	2.4	91km W of Nashik, Maharashtra, India	2021-07-30	23:04:57
2	31.50	74.37	33.0	3.4	49km WSW of Amritsar, Punjab, India	2021-07-30	21:31:10
3	28.34	76.23	5.0	3.1	50km SW of Jhajjar, Haryana	2021-07-30	13:56:31
4	27.09	89.97	10.0	2.1	53km SE of Thimphu, Bhutan	2021-07-30	07:19:38

```
splitted = df['Date'].str.split('-', expand=True)

df['day'] = splitted[2].astype('int')
df['month'] = splitted[1].astype('int')
df['year'] = splitted[0].astype('int')

df.drop('Date', axis=1,
        inplace=True)
df.head()
```

OUTPUT :

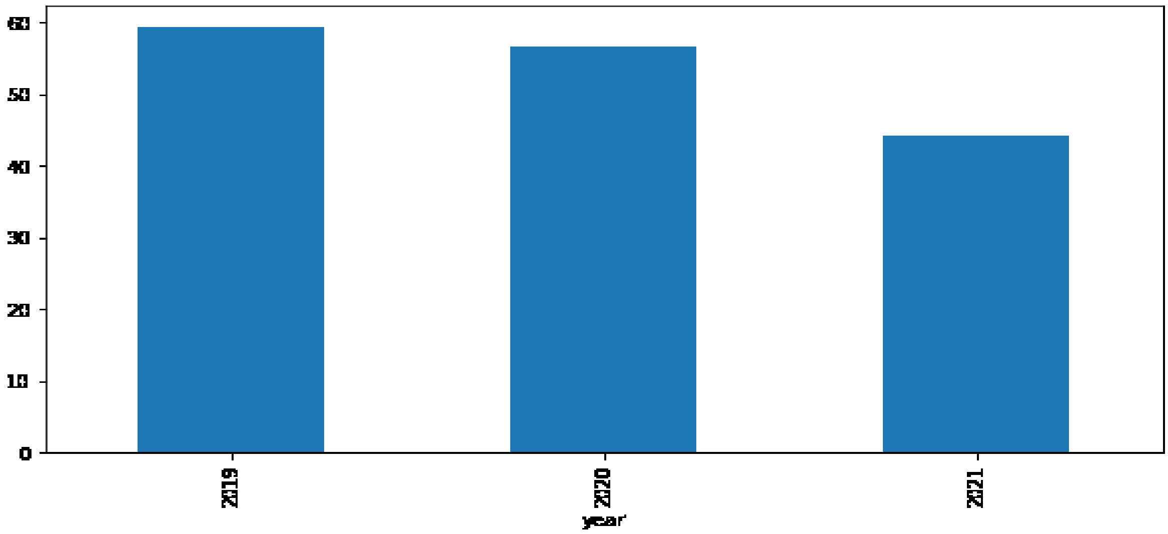
	Latitude	Longitude	Depth	Magnitude	Location	Time	day	month	year
0	29.06	77.42	5.0	2.5	53km NNE of New Delhi, India	09:43:23	31	7	2021
1	19.93	72.92	5.0	2.4	91km W of Nashik, Maharashtra, India	23:04:57	30	7	2021
2	31.50	74.37	33.0	3.4	49km WSW of Amritsar, Punjab, India	21:31:10	30	7	2021
3	28.34	76.23	5.0	3.1	50km SW of Jhajjar, Haryana	13:56:31	30	7	2021
4	27.09	89.97	10.0	2.1	53km SE of Thimphu, Bhutan	07:19:38	30	7	2021

Exploratory Data Analysis :

EDA is an approach to analyzing the data using visual techniques. It is used to discover trends, and patterns, or to check assumptions with the help of statistical summaries and graphical representations.

```
plt.figure(figsize=(10, 5))
x = df.groupby('year').mean()['Depth']
x.plot.bar()
plt.show()
```

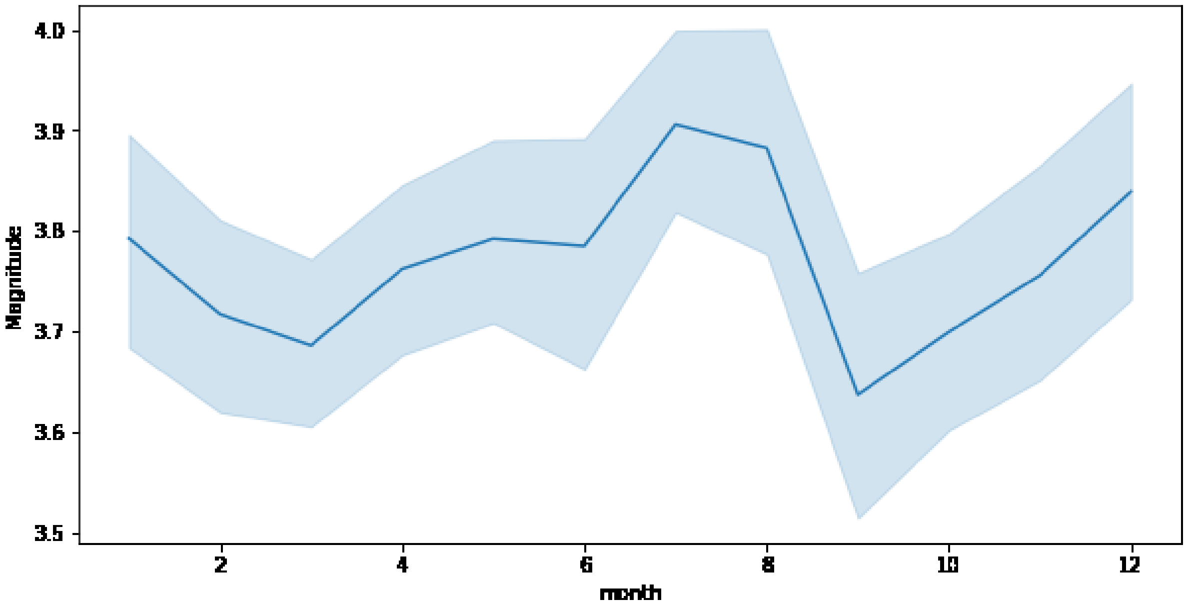
OUTPUT :



The depth from which earthquakes are starting is reducing with every passing year.

```
plt.figure(figsize=(10, 5))
sb.lineplot(data=df,
            x='month',
            y='Magnitude')
plt.show()
```

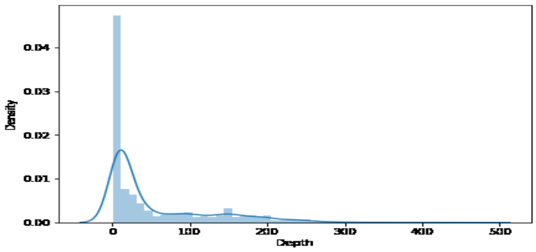
OUTPUT :

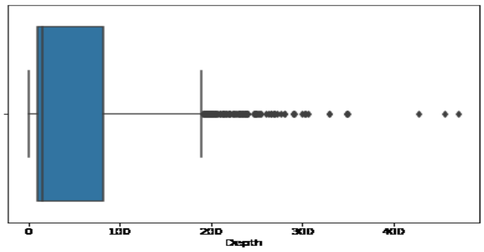


Here we can observe that the changes of an earthquake with higher magnitude are more observed during the season of monsoon.

```
plt.subplots(figsize=(15, 5))

plt.subplot(1, 2, 1)
sb.distplot(df['Depth'])

plt.subplot(1, 2, 2)
sb.boxplot(df['Depth'])

plt.show()
```

OUTPUT :

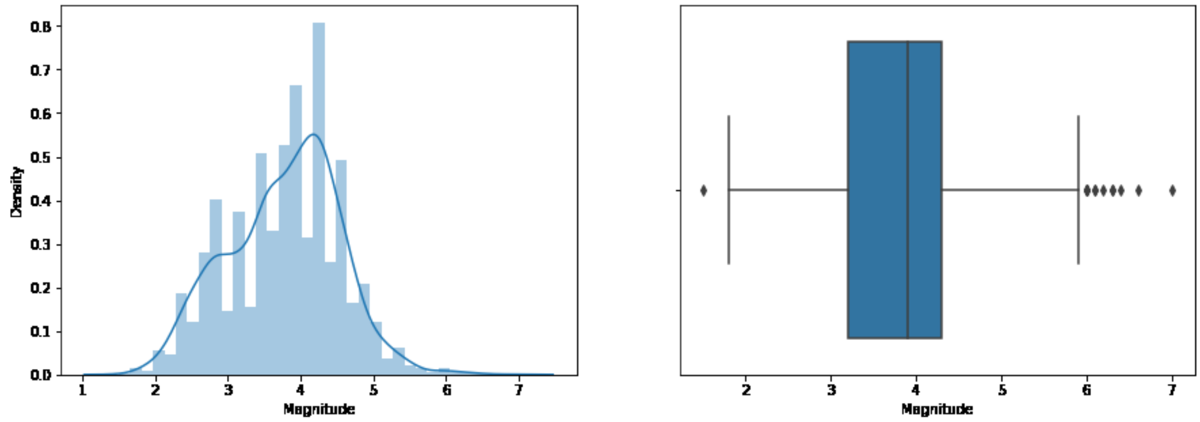




From the distribution graph, it is visible that there are some outliers that can be confirmed by using the boxplot. But the main point to observe here is that the distribution of the depth at which the earthquake rises is left-skewed.

```
plt.subplots(figsize=(15, 5))

plt.subplot(1, 2, 1)
sb.distplot(df['Magnitude'])

plt.subplot(1, 2, 2)
sb.boxplot(df['Magnitude'])

plt.show()
```

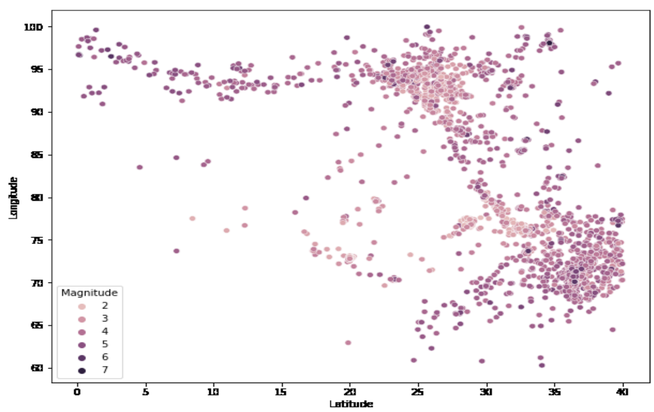
OUTPUT :



As we know that many natural phenomena follow a normal distribution and here we can observe that the magnitude of the earthquake also follows a normal distribution.

```
plt.figure(figsize=(10, 8))
sb.scatterplot(data=df,
               x='Latitude',
               y='Longitude',
               hue='Magnitude')
plt.show()
```

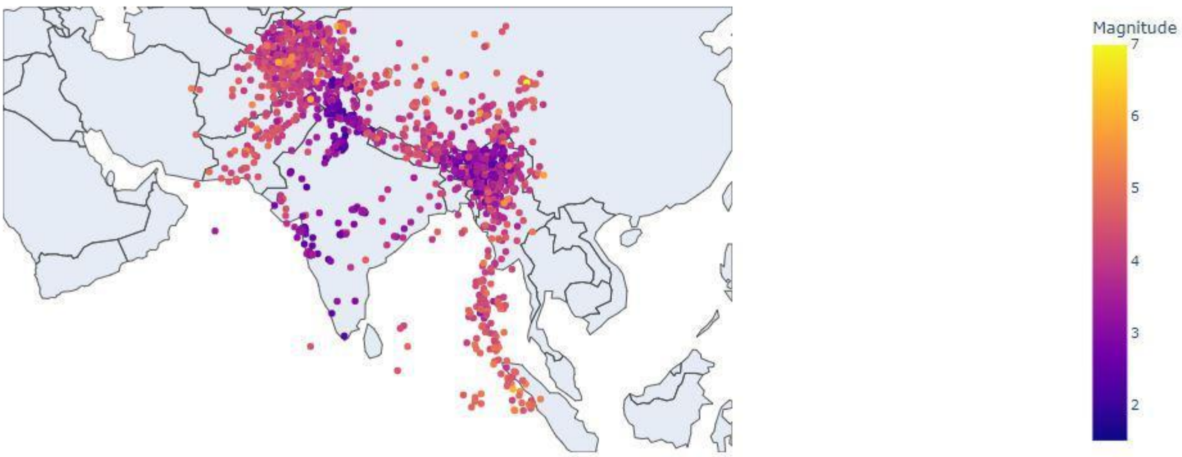
OUTPUT :



Now by using Plotly let's plot the latitude and the longitude data on the map to visualize which areas are more prone to earthquakes.

```
import plotly.express as px
import pandas as pd

fig = px.scatter_geo(df, lat='Latitude',
                     lon='Longitude',
                     color="Magnitude",
                     fitbounds='locations',
                     scope='asia')
fig.show()
```

OUTPUT :



Conclusion:

The earthquake prediction model shows promise in its ability to forecast seismic events. While the current iteration has certain limitations, further enhancements and additional data sources could improve its accuracy. Overall, this project contributes to the ongoing efforts in earthquake prediction research.